

Lecture 4 - Sep 2

Permutations of Multisets

$M = \{n_1 a_1, n_2 a_2, \dots, n_k a_k\}$, when $a_1, \dots, a_k$ are distinct, $n_1, \dots, n_k$ are repetition #'s of $a_1, \dots, a_k$.

Remark: $n_1, \dots, n_k$ can have infinite.

Let $r \in \mathbb{Z}^+$.

An r -permutation of M is an ordered arrangement of r objects in M .

When $r = n_1 + \dots + n_k$, r -permutation is called a *permutation of M* .

Example

$$M = \{3 * a_1, 2 * a_2, \infty * a_4\}$$

1. give a 4-permutation of M

$$a_1 a_2 a_4 a_1, a_4 a_4 a_4 a_4$$

2. 2026-permutation of M :

$$\underbrace{a_4, \dots, a_4}_{2026 \text{ copies}}$$

Derivation

Question: How to determine # r -permutations of M ?

Theorem 2.4.1

Let $M = \{n_1 a_1, \dots, n_k a_k\}$

If $r \leq \min\{n_1, \dots, n_k\}$, then # r -permutations of $M = \underbrace{k * \dots * k}_{r \text{ times}} = \boxed{k^r}$

Theorem 2.4.2

Let $M = \{n_1 * a_1, \dots, n_k a_k\}$

And let $n := n_1 + \dots + n_k$. Then # permutation of $M = \frac{n!}{n_1! n_2! \dots n_k!}$

Scratch work for proof

Consider trying to first place all $n_1 * a_1$. We have n positions to place them in, so we **choose** where to place them (since copies of the same element are not distinguishable from each other):

$$\binom{n}{n_1} * 1$$

After placing them we do the same for $n_2 * a_2$, except now we have n_1 less spots to fill:

$$\binom{n}{n_1} \binom{n-n_1}{n_2} \dots$$

Proof

We have n positions in a row and each position is occupied by one of the n objects. We 1st pick the positions for a_1 , and then the positions for a_2 , and so on.

Thus, for each type there are

$$\binom{n-n_1-n_2-\dots-n_{j-1}}{n_j}$$

ways to place element j , $1 \leq j \leq k$.

By the multiplication principle the # permutations of $M =$

$$\begin{aligned} & \binom{n}{n_1} \binom{n-n_1}{n_2} \dots \binom{n-n_1-n_2-\dots-n_{k-1}}{n_k} \\ &= \frac{n!}{n_1!(n-n_1)!} \frac{(n-n_1)!}{n_2!(n-n_1-n_2)!} \dots \frac{(n-n_1-n_2-\dots-n_{k-1})!}{n_k!(n-n_1-n_2-\dots-n_k)!} \\ &= \frac{n!}{\cancel{n_1!} \cancel{(n-n_1)!}} \frac{\cancel{(n-n_1)!}}{\cancel{n_2!} \cancel{(n-n_1-n_2)!}} \dots \frac{\cancel{(n-n_1-n_2-\dots-n_{k-1})!}}{\cancel{n_k!} \cancel{(n-n_1-n_2-\dots-n_k)!}} \\ &= \frac{n!}{n_1!n_2!\dots n_k!0!} \\ &= \boxed{\frac{n!}{n_1!n_2!\dots n_k!}} \end{aligned}$$

Remarks

1. When $k = 2$, # permutation of $M = \{n_1a_1, n_2a_2\}$

$$\frac{n!}{n_1!n_2!} = \binom{n}{n_1} = \binom{n}{n_2}$$

Question: How many bitstrings of length n are there?

$$2^n = \# \text{ } n\text{-permutation of } M = \{\infty * 0, \infty * 1\}$$

How many bitstrings of length n with exactly k 1's?

$$M = \{(n-k) * 0, k * 1\}$$

$\rightarrow$

$$\binom{n}{k} = \# \text{ permutations of } M.$$

Remark 2: From the proof of [Lecture 4 - Sep 2 > Theorem 2.4.2](#): If we have $n = n_1 + \dots + n_k$ distinct objects and put n_1 of them into Box 1, n_2 of them into Box 2, $\dots$, n_k of them into Box k (boxes are distinguishable)

Remark 3: If boxes are indistinguishable and n_1 is equal to n_k , then there are

Remark 4: if $\min\{n_1, \dots, n_k\} < r < n$, no formula is given. But in chapter 7 we will use generating functions to solve.

Combinations of Multiset

$$M = \{n_1 a_1, \dots, n_k a_k\}, n = n_1 + \dots + n_k$$

An r -combination of M is an unordered selection of r of the objects in M .

Remark: an r -combination of M is an r -element *multisubset* of M .

$$A = \{x_1 a_1 x_2 a_2, \dots\}$$

all combinations must satisfy

$$0 \leq x_1 \leq r, 0 \leq x_2 \leq r, \dots, 0 \leq x_k \leq r$$

$$x_1 + x_2 + \dots + x_k = r$$

1. Each r -combination of M gives an integral solution to $\square$.

2. Each integral solution to $\square$ yields an r -combination of M .

r -combinations of M = # of integral solutions to $\square$.

Question: How to find all integral solutions?

permutations of $M = \{r * \times, (k-1) * 1\}$

$$\binom{r+k-1}{r}$$

This is also called the **stars-and-bars principle**.

We will use the stars-and-bars principle to find all integral solutions.

Theorem 2.5.1

r -combinations of $M = \{n_1 a_1, \dots, n_k a_k\}$ is = # integral solution to $(\square)$.

$$\binom{r+k-1}{r}$$

Proof

Let X be the set of integral solutions to $(\square)$ and Y be the set of permutations of $\{r * \times, (k-1) * 1\}$

It suffices to find a bijection from $X \rightarrow Y$.

For each integral solution $x_1, \dots, x_k$ to $(\square)$:

$$\underbrace{\overbrace{**\dots*}^{x_1} \overbrace{**\dots*}^{x_2} \dots \overbrace{**\dots*}^{x_k}}_{k-1 \text{ bars}}$$

For example consider the examples:

$$\begin{aligned}x_1 + x_2 + x_3 = 5 \quad x_1 = 1, x_2 = 3, x_3 = 1 &\rightarrow *|***|* \\x_1 + x_2 + x_3 = 5 \quad x_1 = 0, x_2 = 0, x_3 = 5 &\rightarrow ||***** \\&\dots\end{aligned}$$

Converly, for each permutation of $\{r*, (k-1)*1\}$

let

$$\begin{aligned}x_1 &= \# \text{ stars to the left of the 1st bar} \\x_2 &= \# \text{ stars between the 1st and 2nd bars} \\&\vdots \\x_k &= \# \text{ stars to the right of the (k-1)th bar}\end{aligned}$$

Then $x_1 + \dots + x_k = r, 0 \leq x_i \leq r$ for all i .

□

REMARKS

1. # nonnegative integral solution to $x_1 + \dots + x_k = r = \binom{r+k-1}{r}$

2. Let

$$\begin{aligned}x_1 + \dots + x_k &= r \\r_1 \leq x_1 \leq r, \dots, r_k \leq x_k \leq r \\&\text{for some } r_1, r_2, \dots, r_k\end{aligned}$$

Can find the number of solutions by using substitution to turn into the normal form:

$$\begin{aligned}y_1 &= x_1 - r_1, y_2 = x_2 - r_2, \dots, y_k = x_k - r_k \\&\rightarrow \\y_1 + y_2 + \dots + y_k &= (x_1 - r_1) + (x_2 - r_2) + \dots + (x_k - r_k) \\&= (x_1 + \dots + x_k) - (r_1 + \dots + r_k)\end{aligned}$$

$$\boxed{y_1 + \dots + y_k = r - (r_1 + \dots + r_k)} \\ \boxed{0 \leq y_1, 0 \leq y_2, \dots, 0 \leq y_k}$$

So the number of solutions becomes:

$$\binom{r - (r_1 + \dots + r_k) + k - 1}{r - (r_1 + \dots + r_k)}$$

3. The conditions from □ *must* hold to use the stars and bars principle. In other words, upper bounds like $x_i < r_i$ invalidate the use of the principle. The inclusion-exclusion principle in chapter 6 will be applied